

Trading Bot Project: Comprehensive Report

1. Project Overview

This project is a high-performance algorithmic trading system designed for XAUUSD (Gold). The system separates the 'Thinking' from the 'Execution' by using a hybrid Python and MetaTrader 5 (MT5) architecture. This ensures maximum flexibility for complex calculations while maintaining the stability of a professional trading terminal.

2. Trade Logic & Strategy

The bot follows a 'Top-Down' approach to find high-probability trades:

- Trend Filter (H4): The bot first checks the 4-hour chart. It only buys in an uptrend and only sells in a downtrend. This ensures we are trading with the 'big money' momentum.
- Market Structure: On the 15-minute chart, it looks for a 'Liquidity Sweep' (where price takes out a recent high or low) followed by a 'Break of Structure' (BOS). This confirms that a reversal is actually happening.
- Entry (Fibonacci): Once a reversal is confirmed, the bot waits for price to 'pull back' to a specific Fibonacci level (usually 0.786). This allows us to enter at a discount price with a very tight Stop Loss.
- Exit Rules: The Stop Loss is placed behind the recent anchor swing, and the Take Profit is set at the recent high/low. This maintains a high reward-to-risk ratio.

3. Execution Details (The Python Bridge)

The 'Brain' of the system is the Python Bridge (ea_bridge.py). Its job is to:

- Monitoring: It polls the MT5 data files every 60 seconds to see if a new signal has developed.
- Signal Persistence: It uses a 'Forward-Fill' technique. Once a signal appears, the bridge keeps it alive for a specific number of bars so that MT5 has enough time to fill the limit order.
- Spam Prevention: It tracks what was previously sent so it doesn't spam the same command over and over, which keeps the system clean and fast.

4. MetaTrader 5 Expert Advisor (The Muscle)

The MT5 EA (myEA.mq5) handles the heavy lifting of actually interacting with the broker:

- Robust I/O: The EA uses special 'File Sharing' settings. This permits Python to read the price data while the EA is writing it, preventing the program from ever hanging or crashing.
- Position Management: Before placing a new trade, the EA checks if there is already an open position. This prevents the bot from accidentally doubling up on the same trade.
- Order Safeties: Every trade command includes an automatic expiration. If the market doesn't hit our entry price within a few hours, the order is cancelled automatically to protect your capital.

5. Optimization (Walk-Forward Validation)

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To ensure the bot works in real-time and isn't just 'lucky' in backtesting, we use Walk-Forward Optimization (WFO):

- Rolling Windows: We split the history into pieces. We train the bot on 6 months of data to find the best settings, then we test it blindly on the next 3 months of unseen data.
- Out-of-Sample Testing: This is the 'gold standard' for trading. It proves that the parameters (like Fib levels) generalize well to new market conditions rather than being over-tuned to the past.
- Metrics: We focus on 'Prop Firm Pass Rate' and 'Return-to-Drawdown' ratios. This ensures the bot is safe enough to handle professional management accounts.

6. Conclusion

By combining Python's advanced math with MT5's reliable execution, this system achieves a professional-grade trading workflow. It is built to be robust, transparent, and most importantly mathematically validated against unseen market data.